# Battery Detection for Robotic E-Waste Extraction

## Introduction

This project focuses on developing a computer vision–based instance segmentation system to detect lithium-ion batteries inside discarded mobile phones for safe robotic extraction in e-waste recycling applications. Accurate battery detection is essential because incorrect extraction may damage the battery, create safety hazards, or reduce robotic efficiency.

The primary objective of this study was to evaluate lightweight YOLO segmentation models for real-time robotic deployment while maintaining high detection accuracy and strong localisation capability.

The evaluated models include:

- YOLOv8
- YOLOv11
- YOLOv26

The evaluation focused on the following requirements:

- High segmentation accuracy
- Reliable battery localisation
- Fast inference speed for robotic integration
- Lightweight architecture for deployment on edge devices
- Strong generalisation performance

### Dataset Overview

The dataset used in this project was a custom annotated battery segmentation dataset developed for robotic e-waste extraction research.

### Dataset Characteristics

| Property | Value |
|---|---|
| Total Images | ~665 |
| Total Battery Instances | ~780 |
| Annotation Type | Instance Segmentation |
| Annotation Tool | CVAT.ai |
| Object Class | Battery |
| Dataset Structure | Train / Validation / Test |

Each image contains a single labelled battery object, resulting in a one-to-one relationship between images and annotations. This simplifies the segmentation task and contributes to the high detection accuracy achieved by the models.

## Key Dataset Observations

- The dataset is relatively small.
- Most batteries are centrally positioned in the images.
- Background complexity is limited.
- Battery appearance variation is moderate.
- Some images were excluded during training due to missing labels or annotation inconsistencies.

The dataset is suitable for YOLO-based segmentation models; however, the limited diversity may reduce generalisation capability in highly complex real-world scenarios.

## Evaluation Metrics

The following performance metrics were used to evaluate the models:

| Metric | Description |
| --- | --- |
| Mask mAP@50 | Segmentation accuracy at IoU threshold 0.50 |
| Mask mAP@50:95 | Average segmentation accuracy across multiple IoU thresholds |
| Precision | Ability to minimise false positive detections |
| Recall | Ability to detect all battery objects |
| Mean Mask IoU | Average overlap between predicted and ground truth masks |
| Latency (ms/image) | Average inference time per image |
| FPS | Frames processed per second |
| Parameters (M) | Total model parameters in millions |
| Centroid Error (px) | Distance between predicted and true object centres |

The centroid localisation error was calculated using the Euclidean distance equation:

$$\boldsymbol{d} = \sqrt{(\boldsymbol{x_2} - \boldsymbol{x_1})^2 + (\boldsymbol{y_2} - \boldsymbol{y_1})^2}$$

-10-8-6-4-2246810-10-5510A(6.0, 6.0)B(-6.0, -6.0)d = 16.97Delta x = 12Delta y = 12

where:

- $(x_1, y_1)$represents the ground truth centroid
- $(x_2, y_2)$represents the predicted centroid

## Experimental Setup

The experiments were conducted using the following environment:

| Component | Specification |
|---|---|
| Platform | Google Colab |
| GPU | NVIDIA Tesla T4 |
| Framework | Python + Ultralytics |
| Task | Instance Segmentation |
| Image Size | 640 × 640 |
| Batch Size | 16 |
| Epochs | 50 |
| Optimisation | Default Ultralytics Settings |

All models were trained using identical hyperparameters to ensure a fair comparison.

# Results and Evaluation

### YOLOv8n-seg

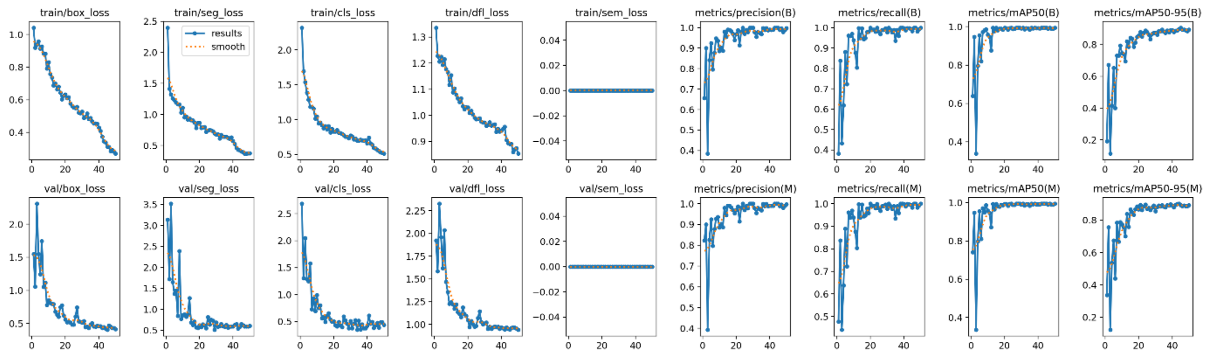


*Figure 1: Training and Validation matrix*

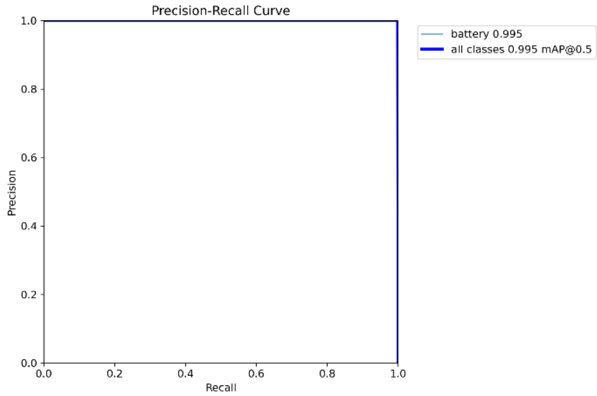


*Figure 2: Precision recall curve*

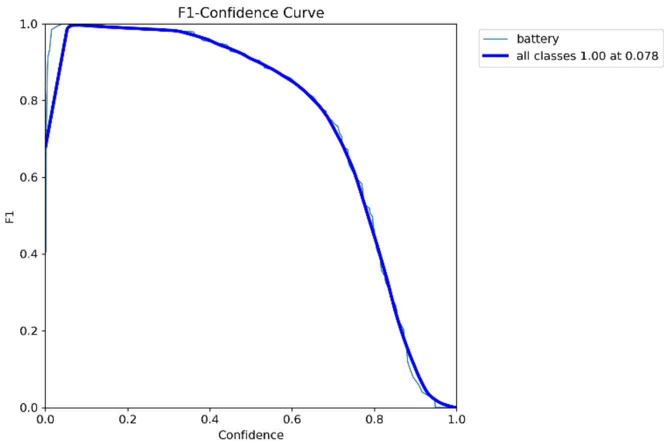


*Figure 3: F1 Score vs Confidence Threshold*

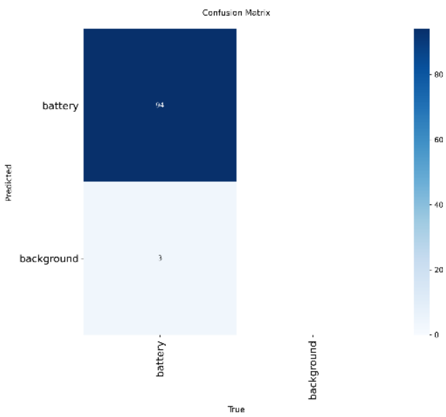


*Figure 4: Confusion Matrix*

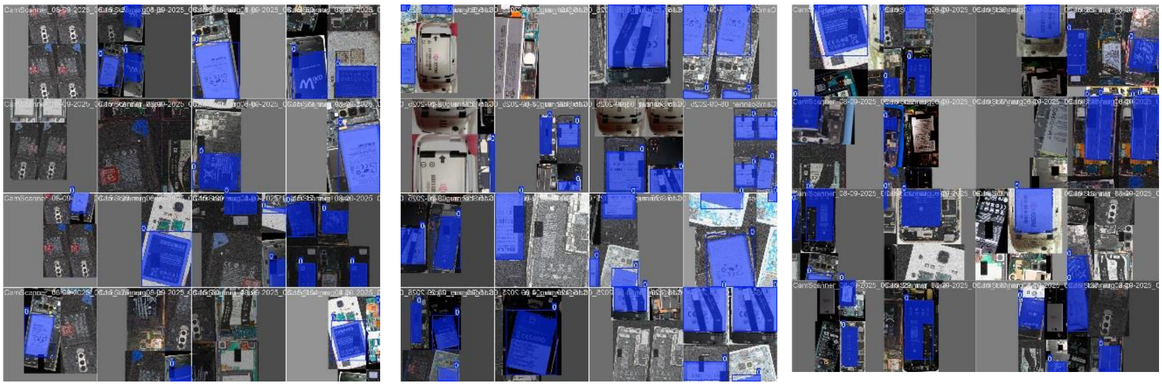

*Figure 5: Train batches*

## YOLOv11n-seg

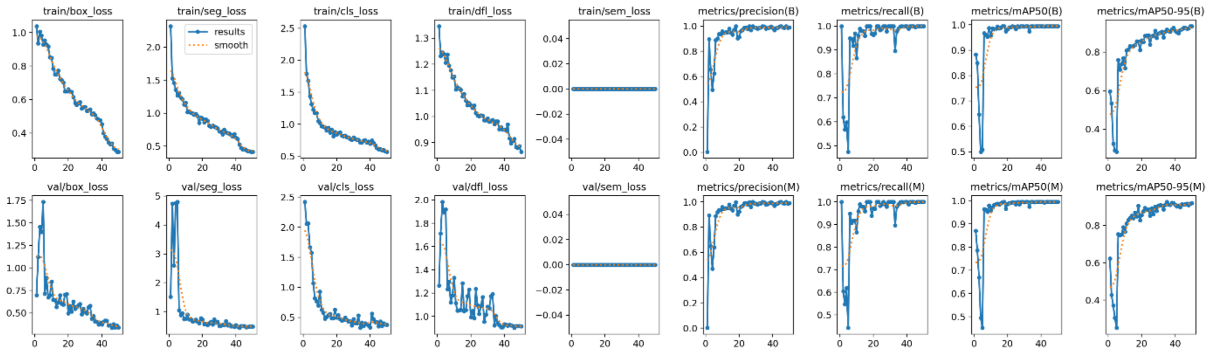


*Figure 6: Training and validation*

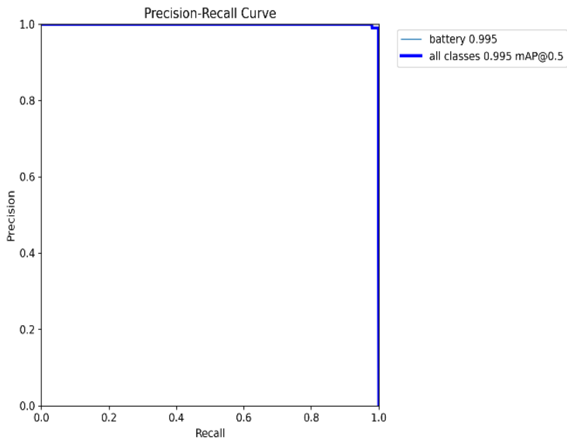


*Figure 7: Precision Recall*

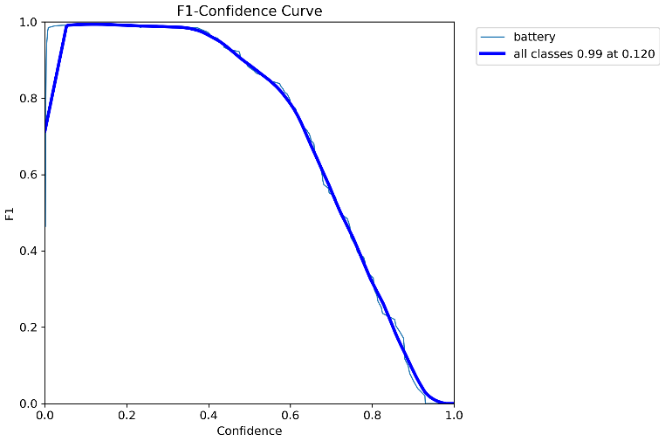


*Figure 8: F1 Score vs Confidence Threshold*

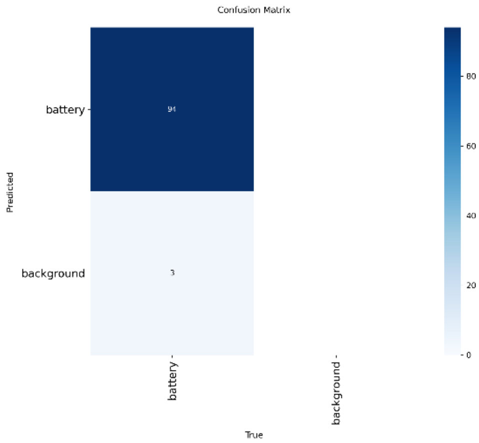


*Figure 9: Confusion Matrix*

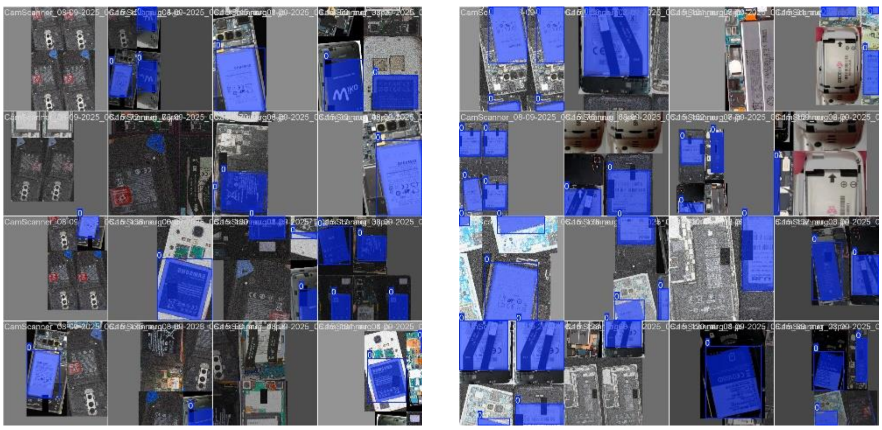

*Figure 10: Training Batches*

## YOLOv26n-seg

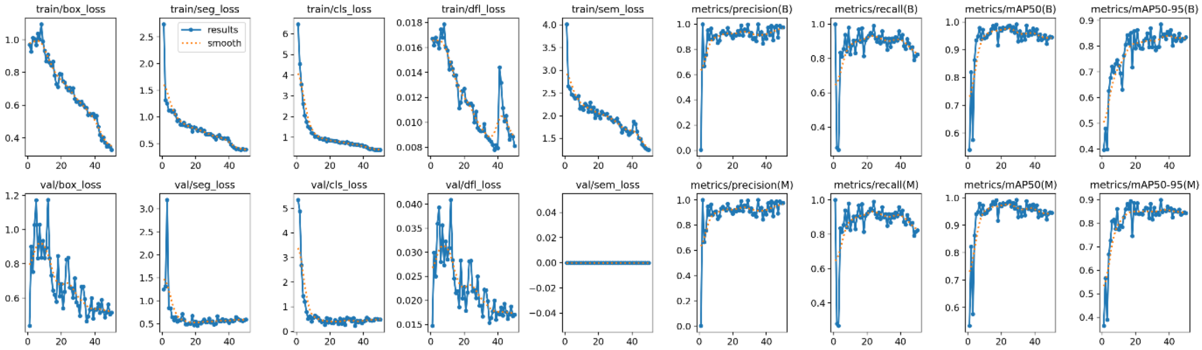


*Figure 11: Training and Validation*

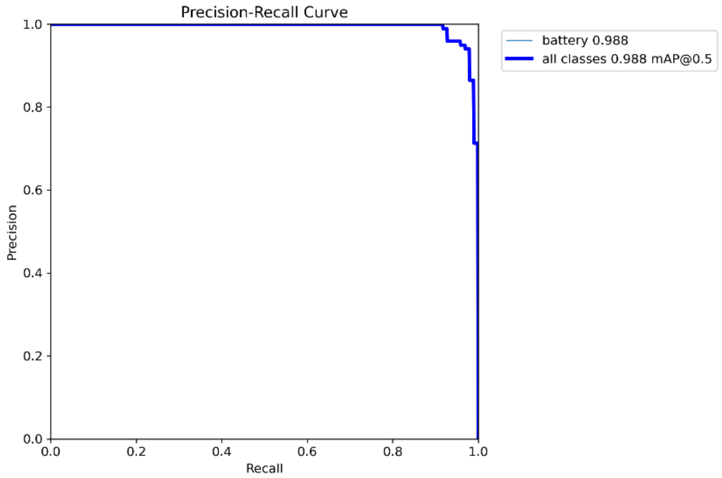


*Figure 12: Precision recall curve*

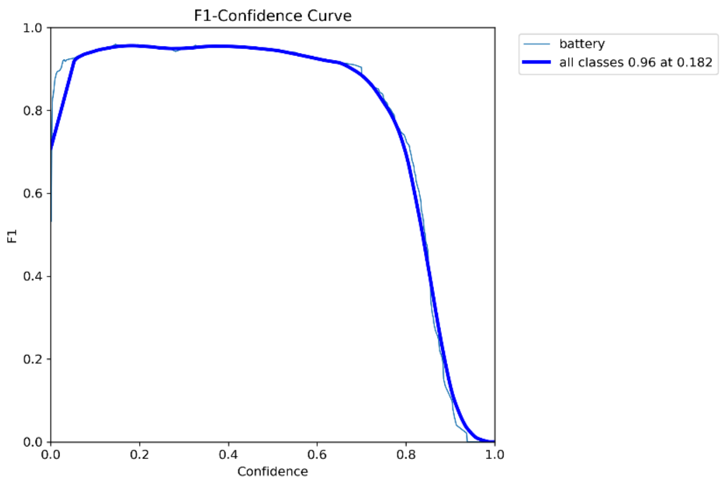


*Figure 13: F1 Score vs Confidence Threshold*

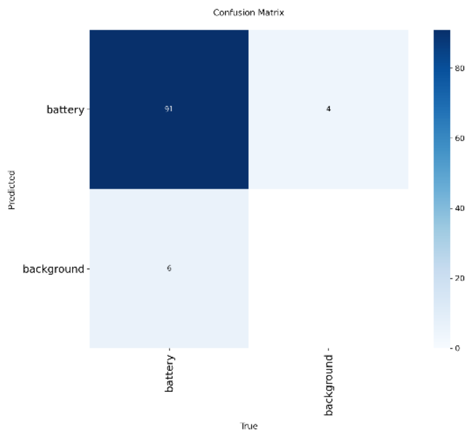


*Figure 14: Confusion Matrix*

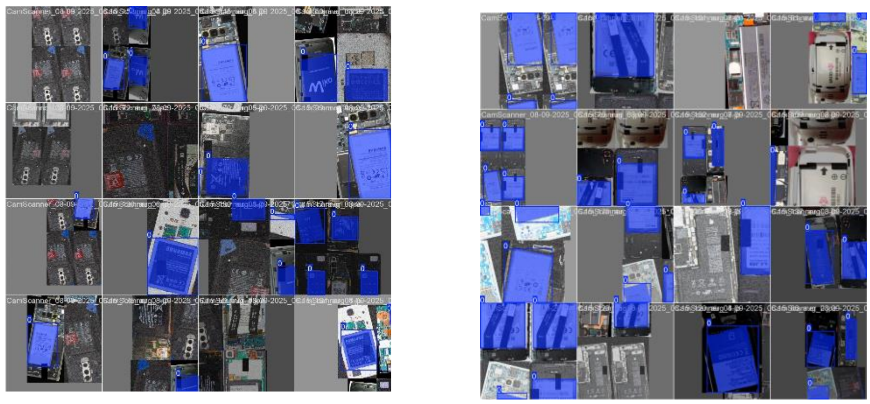

*Figure 15: Training batches*

# Model Performance Analysis

## YOLOv8n-seg (Baseline Lightweight Model)

## Performance

| Metric | Value |
|---|---|
| Mask mAP@50 | 0.9950 |
| Mask mAP@50:95 | 0.9023 |
| Precision | 1.0000 |
| Recall | 0.9942 |
| Mean Mask IoU | 1.0000 |
| Latency (ms/image) | 44.74 |
| FPS | 22.35 |
| Parameters (M) | 3.26 |
| Centroid Error (px) | 0.00 |

### Observations

The YOLOv8n-seg model demonstrated highly stable convergence during training. The box loss, segmentation loss, and classification loss decreased consistently across epochs, indicating effective feature learning and good optimisation behaviour.

Validation losses closely followed the training losses, suggesting minimal overfitting and strong generalisation performance.

The precision–recall behaviour remained highly stable, with near-perfect precision and recall values. The model achieved extremely accurate battery localisation while maintaining lightweight computational requirements.

### Strengths

- Highest Mask mAP@50 score
- Excellent precision and recall
- Strong localisation capability
- Stable training behaviour
- Suitable for real-time deployment

### Limitations

- Slightly slower inference than YOLOv11n and YOLOv26n
- Larger parameter count compared to YOLOv26n

## YOLOv11n-seg

### Performance

| Metric | Value |
|---|---|
| Mask mAP@50 | 0.9948 |
| Mask mAP@50:95 | 0.9200 |
| Precision | 0.9898 |
| Recall | 0.9996 |
| Mean Mask IoU | 1.0000 |
| Latency (ms/image) | 38.40 |
| FPS | 26.04 |
| Parameters (M) | 2.83 |
| Centroid Error (px) | 0.00 |

### Observations

YOLOv11n-seg achieved the highest Mask mAP@50:95 score among all evaluated models, indicating superior segmentation consistency across varying IoU thresholds.

Training and validation losses decreased smoothly without significant fluctuations, showing stable convergence and effective learning behaviour. The precision–recall curve remained close to the ideal top-right region, demonstrating excellent detection robustness.

The model achieved the highest recall value, indicating that it successfully detected nearly all battery objects within the dataset.

### Strengths

- Highest overall segmentation consistency
- Excellent recall performance
- Fast inference speed
- Good balance between speed and accuracy
- Lower parameters count than YOLOv8n

### Limitations

- Slightly lower precision than YOLOv8n
- More computationally demanding than YOLOv26n

## YOLOv26n-seg

### Performance

| Metric | Value |
|---|---|
| Mask mAP@50 | 0.9874 |
| Mask mAP@50:95 | 0.8987 |
| Precision | 1.0000 |
| Recall | 0.9163 |
| Mean Mask IoU | 1.0000 |
| Latency (ms/image) | 35.50 |
| FPS | 28.17 |
| Parameters (M) | 2.69 |
| Centroid Error (px) | 0.00 |

### Observations

YOLOv26n-seg achieved the fastest inference speed and highest FPS among all evaluated models, making it highly suitable for real-time robotic applications.

The model demonstrated effective learning during training; however, minor fluctuations were observed in validation performance compared to YOLOv8n and YOLOv11n, indicating slightly less stable convergence.

Although precision remained extremely high, the lower recall score suggests that some battery objects were occasionally missed during detection.

### Strengths

- Fastest inference speed
- Highest FPS performance
- Smallest parameter count
- Lightweight architecture suitable for edge deployment

### Limitations

- Lower recall than YOLOv8n and YOLOv11n
- Slightly lower segmentation consistency
- Less stable convergence

# Comparative Analysis

## Overall Comparison Table

| Model | Mask mAP@50 | Mask mAP@50:95 | Precision | Recall | Mean Mask IoU | Latency (ms/image) | FPS | Parameters (M) | Centroid Error (px) |
|---|---|---|---|---|---|---|---|---|---|
| YOLOv8n | 0.9950 | 0.9023 | 1.0000 | 0.9942 | 1.0000 | 44.74 | 22.35 | 3.26 | 0.00 |
| YOLOv11n | 0.9948 | 0.9200 | 0.9898 | 0.9996 | 1.0000 | 38.40 | 26.04 | 2.83 | 0.00 |
| YOLOv26n | 0.9874 | 0.8987 | 1.0000 | 0.9163 | 1.0000 | 35.50 | 28.17 | 2.69 | 0.00 |

## Confusion Matrix Insights

The confusion matrices for all YOLO models indicate:

- High true positive detections
- Minimal false positives
- Very low false negative rates
- Strong class separation performance

These results confirm that the models are highly effective at identifying battery objects while minimising incorrect detections.

## Precision–Recall Analysis

Key observations from the precision–recall curves include:

- YOLOv11n achieved the most stable precision–recall behaviour.
- YOLOv8n maintained near-perfect precision across most recall values.
- YOLOv26n showed a slight reduction in recall at higher confidence thresholds.

Overall, the models demonstrated excellent segmentation capability with strong robustness across varying confidence thresholds.

## Speed and Real-Time Performance Analysis

Inference speed is critical for robotic battery extraction systems where real-time decision-making is required.

The evaluation results indicate:

| Model | FPS |
|---|---|
| YOLOv26n | 28.17 |
| YOLOv11n | 26.04 |
| YOLOv8n | 22.35 |

YOLOv26n achieved the best real-time performance due to its lightweight architecture and reduced computational complexity.

## Key Findings

### 1. All YOLO Models Achieved Excellent Segmentation Performance

All evaluated models achieved Mask mAP@50 values above 98%, demonstrating highly accurate battery segmentation capability.

### 2. YOLOv11n Achieved the Best Overall Balance

YOLOv11n demonstrated the strongest balance between:

- Segmentation accuracy
- Recall performance
- Inference speed
- Computational efficiency

Its high recall value makes it highly reliable for robotic extraction tasks where missed detections must be minimised.

#### 3. YOLOv26n Was the Fastest Model

YOLOv26n achieved:

- Lowest latency
- Highest FPS
- Smallest parameter count

This makes it highly suitable for embedded robotic systems with limited hardware resources.

#### 4. YOLOv8n Achieved the Highest Precision

YOLOv8n demonstrated extremely accurate detection performance with perfect precision and very high recall, making it highly reliable for accurate battery segmentation.

## Final Recommendation

### Best Model for Real-Time Deployment

YOLOv11n-seg is recommended for deployment because it provides the best balance between:

- segmentation accuracy,
- recall performance,
- inference speed,
- and computational efficiency.

This makes it highly suitable for robotic e-waste battery extraction systems.

### Best Model for Maximum Speed

YOLOv26n-seg is the best option for applications requiring extremely fast inference and lightweight deployment on edge hardware.

### Best Model for Segmentation Accuracy

YOLOv8n-seg achieved the highest precision and Mask mAP@50 score, making it suitable for applications prioritising segmentation accuracy.

### Future Improvements

The following improvements are recommended for future work:

### Dataset Improvements

- Increase dataset size to improve generalisation
- Include additional phone models
- Introduce varied lighting conditions
- Add complex backgrounds and occlusions
- Reduce positional bias in object placement

### Model Optimisation

- Apply pruning techniques for lightweight deployment
- Use quantisation for embedded hardware optimisation
- Explore synthetic data generation using simulation environments
- Evaluate performance on real robotic hardware systems

## Conclusion

This study evaluated YOLOv8n-seg, YOLOv11n-seg, and YOLOv26n-seg for battery instance segmentation in robotic e-waste extraction applications.

The experimental results demonstrate that all YOLO models achieved exceptionally high segmentation accuracy and localisation performance.

Among the evaluated models:

- YOLOv11n provided the best overall balance between accuracy and efficiency.
- YOLOv26n achieved the fastest real-time inference performance.
- YOLOv8n achieved the highest segmentation precision.

These findings confirm that YOLO-based instance segmentation models are highly suitable for robotic battery extraction systems and can support safe and efficient automation in e-waste recycling applications.

## References

[1] Ultralytics, “Ultralytics Documentation,” 2026. [Online]. Available: https://docs.ultralytics.com. [Accessed: 10-May-2026].

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